

Lecture 7 — Confidence-Aware Revenue Management

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7.1 Overview

Lecture 6 gave the fluid LP two interpretations:

dual prices	:	approximate opportunity costs,
primal counts	:	predicted offline action counts.

The fluid plan is a point prediction based on mean future demand. The central question is:

How much should we trust the fluid prediction?
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The main path is

fluid target $\longrightarrow \sqrt{T}$ uncertainty $\longrightarrow$ confidence region $\longrightarrow$ safe offline counts.
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The central lesson is not that the fluid LP is a bad approximation. It is that a mean prediction should be accompanied by a measure of uncertainty.

7.2 A two-value online allocation problem

There are T arrivals and capacity

$$B = T/2,$$

where T is even. Each arrival is independently

type	H	L
probability	1/2	1/2
reward	2	1.

Every accepted request consumes one unit. Let Z_H be the total number of high-value arrivals.

The time-zero fluid LP is

$$\begin{aligned} \max \quad & 2x_H + x_L \\ \text{s.t.} \quad & x_H + x_L \leq B, \\ & x_H \leq T/2, \\ & x_L \leq T/2, \\ & x_H, x_L \geq 0. \end{aligned}$$

Its solution is

$$x_H^* = B, \quad x_L^* = 0.$$

The fluid model predicts exactly enough high-value demand to fill capacity. A corresponding static policy accepts every H and rejects every L .

7.3 What does the clairvoyant do?

The offline optimum knows Z_H .

- If $Z_H \geq B$, it accepts B high requests and earns $2B$.
- If $Z_H < B$, it accepts all Z_H high requests and fills the remaining $B - Z_H$ units with low requests.

Thus

$$\text{OPT} = 2 \min\{Z_H, B\} + (B - Z_H)^+.$$

The static high-only policy earns

$$2 \min\{Z_H, B\}.$$

Its sample-path loss is therefore

$$\text{OPT} - \text{static reward} = (B - Z_H)^+. \quad (7.1)$$

Because $Z_H \sim \text{Binomial}(T, 1/2)$,

$$\mathbb{E}[Z_H] = B, \quad \text{sd}(Z_H) = \frac{\sqrt{T}}{2}.$$

A central-limit or Berry–Esseen argument then gives

$$\mathbb{E}[(B - Z_H)^+] = \Theta(\sqrt{T}). \quad (7.2)$$

The mean is correct at scale T but uncertain at scale $\sqrt{T}$.

7.4 Three performance scales

The example separates three different performance scales:

$$\boxed{\underbrace{\Theta(T)}_{\text{first-order value}} \quad \underbrace{\Theta(\sqrt{T})}_{\text{stochastic fluctuation}} \quad \underbrace{O(1)}_{\text{strong additive target}} .}$$

A fluid approximation captures the first-order scale. Concentration identifies the fluctuation scale. The strongest policies ask whether prediction and adaptation can reduce the remaining comparison with hindsight to a bounded additive quantity.

7.4.1 Ratio optimality versus additive loss

The offline optimum earns order T , while the static policy loses order $\sqrt{T}$. Therefore

$$\frac{\mathbb{E}[\text{OPT} - \text{static reward}]}{\mathbb{E}[\text{OPT}]} = O(T^{-1/2}) \rightarrow 0.$$

The policy is asymptotically optimal as a percentage of revenue, yet its additive loss diverges. This distinction matters when the allocation problem is repeated over many flights, markets, campaigns, or inventory pools. A vanishing percentage loss can coexist with a meaningful dollar loss on every instance.

7.5 Why not use a fixed hedge?

The high-only policy protects all B units for high types. A natural hedge admits some low types. But any fixed protection decision faces two opposing errors:

- if high demand is below its mean, rejecting too many lows leaves capacity unused;
- if high demand is above its mean, accepting too many lows displaces highs.

Both deviations have scale $\sqrt{T}$. Before arrivals are observed, a static policy cannot know which tail will occur.

This is the same statistical phenomenon as a confidence interval:

$$Z_H = \frac{T}{2} \pm O(\sqrt{T}).$$

The fluid mean is a point estimate; a control policy needs to know how much uncertainty surrounds it.

7.6 Re-solving reacts, but still predicts

With τ periods and b units remaining, the re-solving fluid LP is

$$\begin{aligned} \max \quad & 2x_H + x_L \\ \text{s.t.} \quad & x_H + x_L \leq b, \\ & x_H \leq \tau/2, \\ & x_L \leq \tau/2, \\ & x_H, x_L \geq 0. \end{aligned}$$

Re-solving responds to realized scarcity. If remaining capacity is large relative to expected future high demand, it becomes more willing to accept low types.

But it still uses the point prediction

$$Z_H(\tau) \approx \tau/2.$$

The question remains: when is an action suggested by this prediction safe enough to take?

7.7 From a point forecast to a confidence interval

Hoeffding's inequality gives, for any $u > 0$,

$$\mathbb{P}(|Z_H(\tau) - \tau/2| > u) \leq 2 \exp\left(-\frac{2u^2}{\tau}\right). \quad (7.3)$$

Choosing, for example,

$$u(\tau) = \sqrt{2\tau \log \tau}$$

makes the failure probability polynomially small in τ .

Instead of predicting only

$$Z_H(\tau) \approx \tau/2,$$

we now predict a range:

$$Z_H(\tau) \in \left[\frac{\tau}{2} - u(\tau), \frac{\tau}{2} + u(\tau)\right]$$

with high probability.

This is directly analogous to Lecture 4. UCB placed a confidence interval around an unknown arm mean. Here we place a confidence interval around an unknown future arrival count.

7.8 Safe counts

Suppose the fluid solution predicts that the offline optimum will use x^{fl} units of some action. If the prediction may be wrong by at most u , then only

$$L = [x^{\text{fl}} - u]^+ \quad (7.4)$$

units should be regarded as certainly safe.

$$\text{safe count} = \text{fluid predicted count} - \text{confidence buffer.}$$

For example, if the fluid LP predicts that hindsight will want 100 units of an action and the uncertainty buffer is 12, then 88 units can be treated as a conservative action budget. The remaining 12 are not necessarily wrong; they are simply not certified by the confidence argument.

Remark 7.1 (Why this is not a safety stock in disguise). *A traditional safety stock protects physical inventory against demand uncertainty. Here the buffer protects a prediction of an offline action count. The objects are related, but the latter is designed to support a comparison with a clairvoyant benchmark.*

7.9 A gentle network generalization

With n arrival types, the future is the vector

$$Z(\tau) = (Z_1(\tau), \dots, Z_n(\tau)), \quad \mathbb{E}[Z(\tau)] = \tau\lambda.$$

A union bound and Hoeffding's inequality give

$$\mathbb{P}(\|Z(\tau) - \tau\lambda\|_{\infty} > u) \leq 2n \exp\left(-\frac{2u^2}{\tau}\right). \quad (7.5)$$

Solve the fluid LP at expected future counts and let

$$x^{\text{fl}}(\tau)$$

be a selected optimal action-count vector. The missing link is stability.

Proposition 1 (Informal LP stability). *For a fixed finite-dimensional packing problem, suppose optimal solutions can be selected so that changing future type counts by at most u changes the selected optimal action counts by at most κu , where κ is an instance-dependent constant. Then*

$$\|Z(\tau) - \tau\lambda\|_{\infty} \leq u$$

implies

$$\|x^{\text{off}}(Z(\tau)) - x^{\text{fl}}(\tau)\|_{\infty} \leq \kappa u.$$

This motivates the network safe budget

$$L_{ja}(\tau) = [x_{ja}^{\text{fl}}(\tau) - \kappa u(\tau)]^+. \quad (7.6)$$

If $L_{ja}(\tau) > 0$, then on the confidence event some suitable offline optimum still has room for action a on type j .

The exact stability statement requires care about degeneracy and tie-breaking. Its economic content is that a stable optimization converts confidence about demand into confidence about hindsight's actions.

7.10 From safe counts to online actions

We began with a fluid prediction and added uncertainty:

mean future demand $\rightarrow$ fluid action counts $\rightarrow$ confidence buffer $\rightarrow$ safe offline counts.

But we have not yet answered two questions:

1. Why does agreeing with a predicted offline action control regret?
2. How should an online policy choose among actions using the safe budgets?

Lecture 8 answers both using compensated coupling and the Bayes Selector.

7.11 Takeaways

1. A fluid plan can be accurate at scale T and still miss additive revenue by order $\sqrt{T}$.
2. The scales T , $\sqrt{T}$, and 1 distinguish first-order value, stochastic fluctuations, and a strong additive target.
3. Re-solving adapts to the state but still produces a point prediction.
4. Confidence intervals quantify uncertainty in future arrival counts.
5. LP stability transfers demand uncertainty into uncertainty about offline action counts.
6. A safe count subtracts a confidence buffer from the fluid prediction.
7. The next step is to turn confidence-aware predictions into online actions and a regret bound.

Suggested reading

Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, for the offline-prediction viewpoint. Banerjee and Freund, *Good Prophets Know When the End Is Near*, Sections 1–3, develops the direct confidence-bound interpretation: fluid action counts minus a concentration buffer form safe lower bounds on offline actions.